

2015 HKDSE Physics

Paper 2 Elective Sections

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Section A Astronomy and Space Science

Question 1.1

A satellite is orbiting the Earth at a distance h from the Earth's surface. What is the gain in gravitational potential energy of the satellite in the orbit with respect to the Earth's surface?

m = mass of the satellite

R = radius of the Earth

g = acceleration due to gravity on the Earth's surface

A. $mgh[R/(R + h)]$

B. $mgh[R/(R + h)]^2$

C. $mgh[(R + h)/R]$

D. $mgh[(R + h)/R]^2$

Official answer: A

Use $U = -GMm/r$, with distance measured from the Earth's centre. $\Delta U = GMm[1/R - 1/(R + h)] = GMmh/[R(R + h)]$. Since $g = GM/R^2$ at the surface, $\Delta U = mghR/(R + h)$. The approximation mgh alone assumes h is small compared with R .

Question 1.2

Where is the best location on Earth to build an observatory so as to observe most of the celestial sphere?

A. latitude 90° N

B. latitude 90° S

C. latitude 0°

D. It is the same for any latitude.

Official answer: C

At the equator, the celestial poles lie on the horizon and stars from both celestial hemispheres rise above the horizon as Earth rotates. Over suitable observing times this provides access to the greatest fraction of the celestial sphere. At either pole only one celestial hemisphere is visible.

Question 1.3

The Earth receives solar radiation of power P_0 per unit area. Estimate the power of solar radiation received per unit area on Pluto which is 40 AU from the Sun.

- A. $P_0/39$
- B. $P_0/40$
- C. $P_0(1/39)^2$
- D. $P_0(1/40)^2$

Official answer: D

Radiation spreads over a sphere of area $4\pi r^2$, so received power per unit area varies as $1/r^2$. Earth is 1 AU from the Sun and Pluto is 40 AU away. Thus the value at Pluto is $P_0(1/40)^2 = P_0/1600$. Use distance from the Sun, not the Earth–Pluto separation.

Question 1.4

Which of the following observations by Galileo contradict with the geocentric model of the universe?

- (1) the discovery of satellites of Jupiter
- (2) the retrograde motion of Mars
- (3) the changing phase of Venus

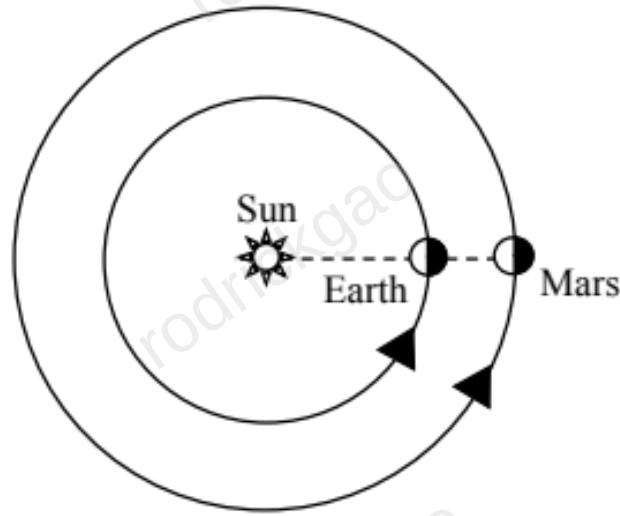
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

Jupiter's satellites show that not all celestial bodies orbit Earth. The full sequence of Venus's phases is inconsistent with the traditional geocentric arrangement and supports Venus orbiting the Sun. Retrograde motion alone is not decisive because the geocentric model used epicycles to account for it. Thus (1) and (3).

Question 1.5

When the Earth lines up with the Sun and Mars as shown, how does Mars appear to move across the night sky as viewed from Earth?



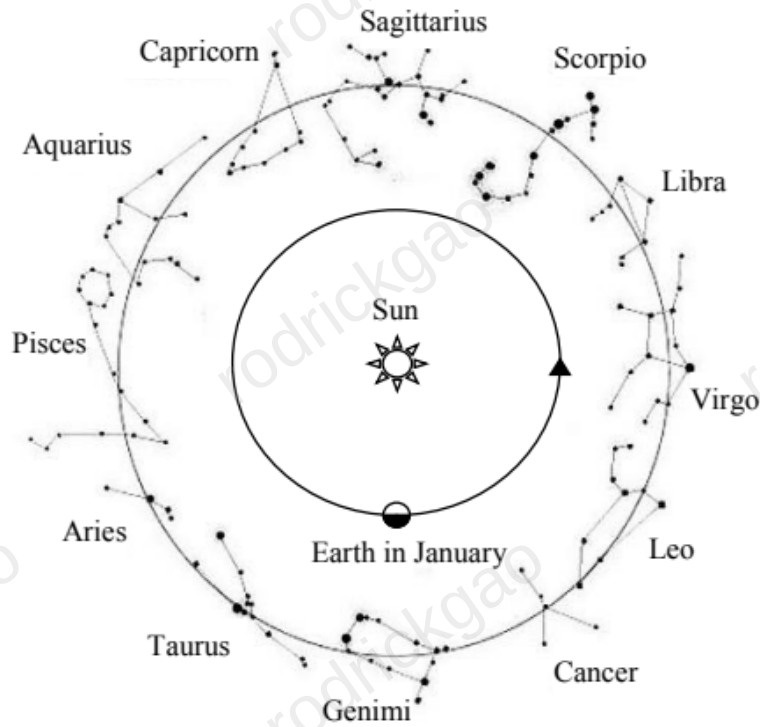
- A. Mars moves from west to east against the background stars.
- B. Mars moves from east to west against the background stars.**
- C. Mars does not move against the background stars.
- D. The movement of Mars cannot be determined because the east and west directions are not known.

Official answer: B

At opposition, Earth overtakes Mars on its faster inner orbit. The changing line of sight makes Mars show apparent retrograde motion: east to west against the background stars. This is the motion relative to the stars, not simply the daily motion of the whole sky.

Question 1.6

Which constellations and in what sequence can be seen to pass the Earth's meridian at night in January?



- A. Capricorn, Sagittarius, Scorpio
- B. Scorpio, Sagittarius, Capricorn
- C. Taurus, Gemini, Cancer
- D. Cancer, Gemini, Taurus

Official answer: C

The January night side of Earth faces away from the Sun toward Taurus, Gemini and Cancer in the diagram. Earth's rotation brings them across the meridian in the order Taurus → Gemini → Cancer. Capricorn, Sagittarius and Scorpio lie toward the daytime side.

Question 1.7

Star	Absolute magnitude	Apparent magnitude
X	2.8	4.7
Y	5.4	3.2

According to the information given above, which of the following about stars X and Y is/are correct?

- (1) Luminosity of star X is greater than that of star Y.
 (2) A telescope collects more energy per unit area per unit time from star X than from star Y.

A. Only (1) is correct.

B. Only (2) is correct.

C. Both (1) and (2) are correct.

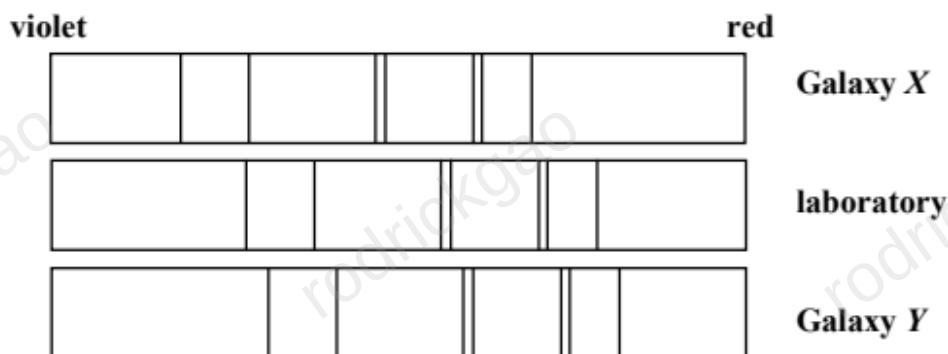
D. Both (1) and (2) are incorrect.

Official answer: A

A smaller absolute magnitude means greater intrinsic luminosity: $2.8 < 5.4$, so X is more luminous. A smaller apparent magnitude means greater received brightness: Y has 3.2 compared with X's 4.7, so the telescope receives more energy per unit area per unit time from Y. Only (1) is correct.

Question 1.8

The respective absorption spectra of hydrogen from Galaxy X, in the laboratory, and Galaxy Y are shown below:



Which of the following descriptions about the motions of Galaxy X and Galaxy Y and their velocities v_X and v_Y relative to the Earth is correct?

A. X: moving away from Earth; $|v_X| < |v_Y|$; Y: moving towards Earth

B. X: moving away from Earth; $|v_X| > |v_Y|$; Y: moving towards Earth

C. X: moving towards Earth; $|v_X| < |v_Y|$; Y: moving away from Earth

D. X: moving towards Earth; $|v_X| > |v_Y|$; Y: moving away from Earth

Official answer: D

X's absorption lines shift toward violet (shorter wavelength), so X approaches Earth. Y's shift toward red, so Y recedes. The shifts of corresponding lines are larger for X; since $|v_r|/c \approx |\Delta\lambda|/\lambda_0$, $|v_X| > |v_Y|$. Hence D.

Question 1 Structured Question

Figure 1.1 shows a distant binary star system viewed by an observer on Earth who is also on the orbital plane of the two stars. The system consists of stars 1 and 2 with masses m_1 and m_2 respectively orbiting in uniform circular motion about their centre of mass O under their mutual gravitational force. They move with the same period in two orbits of radii r_1 and r_2 with orbital speeds v_1 and v_2 respectively.

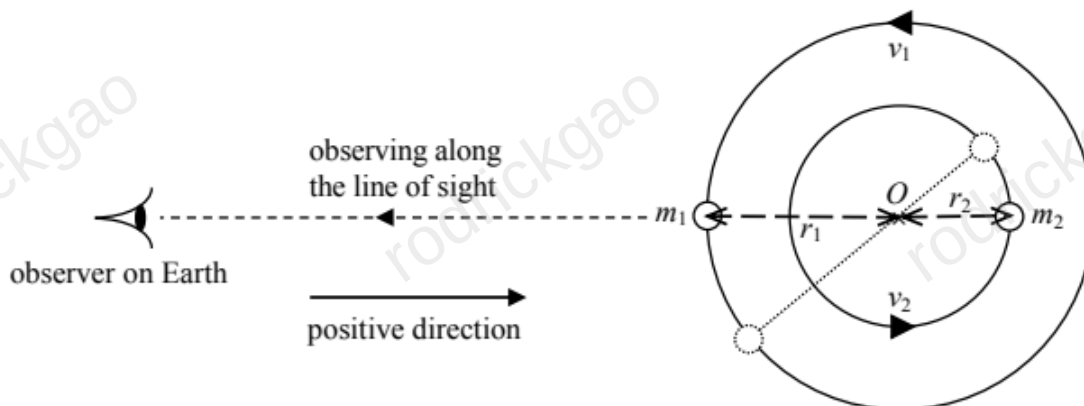


Figure 1.1

By finding the radial velocity v_r of the two stars inferred from the Doppler shift ($\Delta\lambda$) of the hydrogen-alpha line (H_α) observed on Earth, astronomers are able to deduce the masses of the stars. Assume that the centre of mass O of the binary system is stationary with respect to the observer. Figure 1.2 shows the radial velocity curves for the two stars. The direction moving away from the observer is taken to be positive velocity.

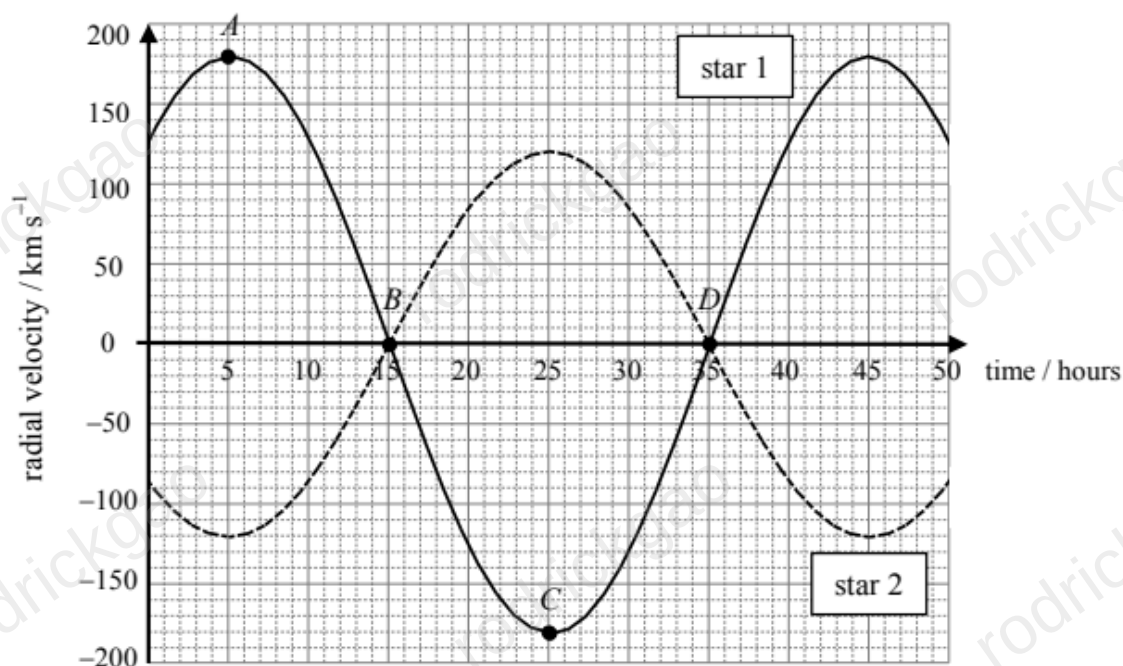


Figure 1.2

- (a)(i) What does it mean by radial velocity v_r of a star observed on Earth? (1 mark)
- (a)(ii) Identify which point, A, B, C or D, marked on the radial velocity curve corresponds to the orbital position of star 1 (in solid line) at the instant shown in Figure 1.1. (1 mark)
- (b) Find, from Figure 1.2, the orbital speed v_1 of star 1 and calculate its orbital radius r_1 . Using a similar method, or otherwise, find the orbital radius r_2 of star 2. (4 marks)

(c) By considering the circular motion of star 1, calculate the mass m_2 of star 2. (2 marks)

(d) A spectrometer can only measure change of wavelength larger than 0.5 nm. Explain whether this instrument is suitable to measure the Doppler shift $\Delta\lambda$ of the hydrogen-alpha line ($\lambda_0 = 656.28$ nm) of the two stars. (2 marks)

Official marking scheme

Solution		Marks	Remarks
1.	(a) (i) radial velocity is the component of the star's velocity along the observer's line of sight / velocity along the observer direction or the line joining the star and the observer.	1A	
		1	
	(ii) Point D	1A	
		1	
	(b) $v_1 = 180 \text{ km s}^{-1}$	1A	
	$v_1 = \frac{2\pi r_1}{T} = \frac{2\pi r_1}{40 \times 60 \times 60}$ (Period $T = 40$ hr)	1M	
	$r_1 = 4.125 \times 10^6 \text{ km}$ or $4.125 \times 10^9 \text{ m}$	1A	
	From figure, $v_2 = 120 \text{ km s}^{-1}$; and by ratio or similar calculation gives	1A	
	$r_2 = 2.75 \times 10^6 \text{ km}$ or $2.75 \times 10^9 \text{ m}$	4	
	(c) $\frac{Gm_1m_2}{(r_1+r_2)^2} = m_1\left(\frac{2\pi}{T}\right)^2 r_1 = \frac{m_1 v_1^2}{r_1} \left[\omega = \frac{2\pi}{T}\right]$	1M	
	$\frac{(6.67 \times 10^{-11}) m_2}{(4.125 \times 10^9 + 2.75 \times 10^9)^2} = \frac{(180 \times 10^3)^2}{4.125 \times 10^9}$	1A	
	Therefore, $m_2 = 5.57 \times 10^{30} \text{ kg}$	2	
	(d) $\frac{v_r}{c} = \frac{\Delta\lambda}{\lambda} = \frac{0.5 \text{ nm}}{656.28 \text{ nm}} \Rightarrow v_r = 228.3 \text{ km s}^{-1} > 180 \text{ km s}^{-1}$;	1M	
	Or $\frac{\Delta\lambda}{\lambda} = \frac{v_r}{c} = \frac{180 \times 10^3}{3 \times 10^8} \Rightarrow \Delta\lambda = 0.394 \text{ nm} < 0.5 \text{ nm}$;	1M	
	therefore NOT suitable. Accept using 120 km s^{-1} , $\Delta\lambda = 0.263 \text{ nm} < 0.5 \text{ nm}$	1A	
		2	

(a)(i) Radial velocity is the component of the star's velocity along the observer's line of sight, relative to the observer.

(a)(ii) D. At the illustrated leftmost orbital position, star 1 moves perpendicular to the line of sight, so $v_r = 0$. Immediately afterwards it moves away from the observer, so v_r changes from negative to positive.

(b) Because the observer lies in the orbital plane, the radial-velocity amplitude equals orbital speed: $v_1 = 180 \text{ km s}^{-1}$, $v_2 = 120 \text{ km s}^{-1}$. The period is $40 \text{ h} = 144000 \text{ s}$.

(b) From $v = 2\pi r/T$, $r_1 = (180 \times 10^3)(144000)/(2\pi) \approx 4.13 \times 10^9 \text{ m}$ and $r_2 = (120 \times 10^3)(144000)/(2\pi) \approx 2.75 \times 10^9 \text{ m}$.

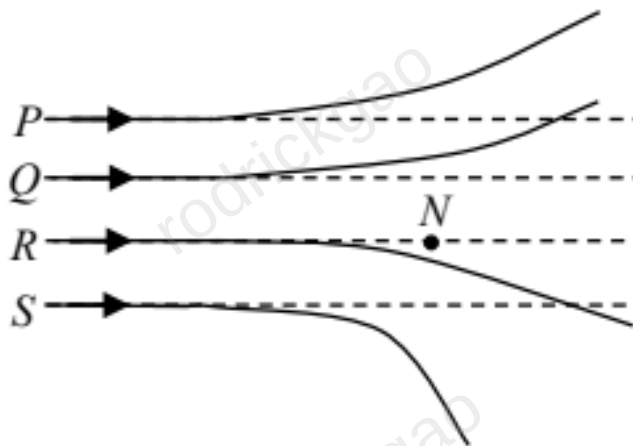
(c) The gravitational separation is $r_1 + r_2$, whereas star 1's circular-orbit radius is r_1 . Thus $Gm_1m_2/(r_1 + r_2)^2 = m_1v_1^2/r_1$. Rearranging, $m_2 = v_1^2(r_1 + r_2)^2/(Gr_1) \approx 5.57 \times 10^{30} \text{ kg}$.

(d) The maximum wavelength shifts are $\lambda_0 v_1/c = 0.394 \text{ nm}$ and $\lambda_0 v_2/c = 0.263 \text{ nm}$, both below 0.5 nm . The instrument is not suitable for either star. Equivalently, its detectable radial speed must exceed about 229 km s^{-1} , larger than either orbital speed.

Section B Atomic World

Question 2.1

A beam of α -particles with the same initial kinetic energy are scattered by a heavy nucleus N. In the figure, if P is a possible path for one of the α -particles, which of the paths, Q, R and S, is/are possible for these α -particles?



- A. Q and R only
- B. R and S only
- C. Q only
- D. S only

Official answer: D

The positively charged nucleus repels α -particles. A smaller impact parameter gives greater deflection for the same initial kinetic energy. Q passes closer than P but is drawn less deflected, so it is impossible. R is aimed directly at the nucleus and should reverse rather than bend sideways as drawn. S is deflected away from the nucleus more strongly and is possible.

Question 2.2

Which of the following provides experimental evidence for discrete energy levels in atoms?

- (1) the spectrum of a sodium discharge tube
- (2) the spectrum of a tungsten filament lamp
- (3) the diffraction of electrons by atomic spacing in crystals

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

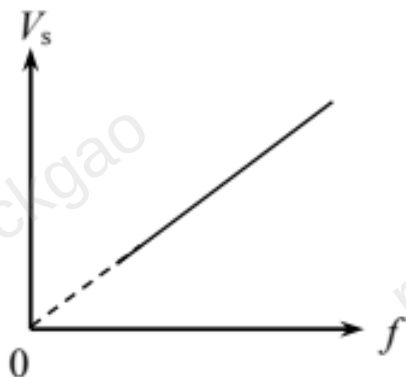
Official answer: A

The sodium discharge tube gives a line spectrum: only particular photon energies occur, showing discrete differences between atomic energy levels. A hot tungsten filament gives a continuous thermal spectrum. Electron diffraction demonstrates matter-wave behaviour, not directly discrete atomic energy levels. Only (1).

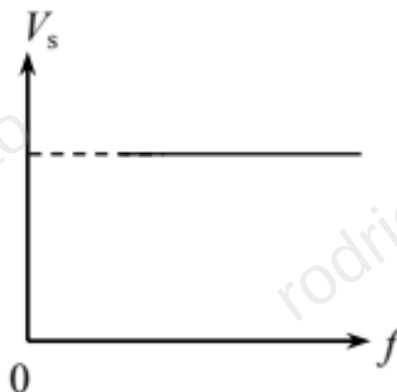
Question 2.3

In a photoelectric experiment, monochromatic light of frequency f is incident upon a metal surface and the stopping potential for the photoelectrons emitted is V_s . If the frequency f is changed, how would V_s vary with f ?

A.



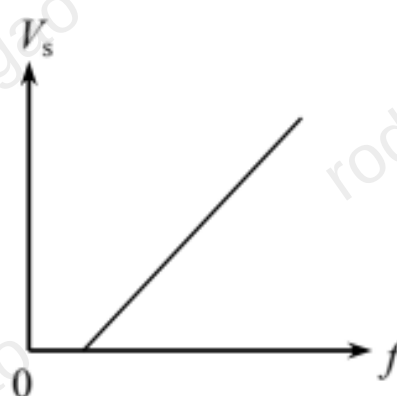
B.



C.



D.

**Official answer: D**

Einstein's equation gives $eV_s = hf - \phi$, hence $V_s = (h/e)f - \phi/e$. Above threshold it is a straight line of positive slope, meeting the frequency axis at $f_0 = \phi/h > 0$. There are no emitted photoelectrons below threshold. This is graph D.

Question 2.4

A spy aircraft is cruising at a height of 10 km above the Earth's surface. It is equipped with a camera having an objective lens of aperture 10 cm. Estimate the minimum separation of two small objects on the Earth's surface to be distinguished by this camera. Assume that both objects emit light of wavelength 500 nm.

A. 0.05 m

B. 0.061 m

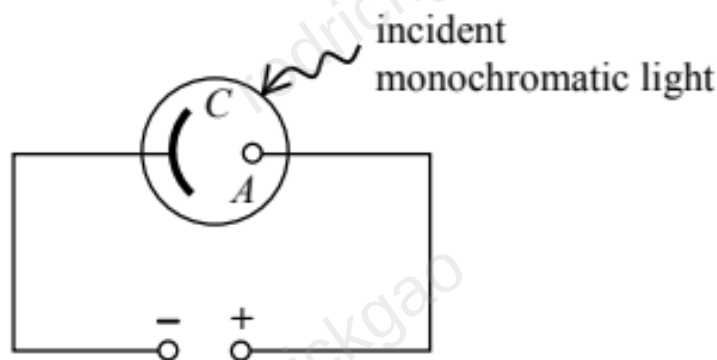
C. 0.10 m

D. 0.122 m

Official answer: B

For a circular aperture, $\theta_{\min} = 1.22\lambda/d = 1.22(500 \times 10^{-9})/0.10 = 6.10 \times 10^{-6}$ rad. At distance 10^4 m, the minimum separation is $x \approx L\theta_{\min} = 0.061$ m. Include the circular-aperture factor 1.22.

Question 2.5



A photocell is connected to a d.c. source as shown. Monochromatic light is incident on cathode C of the photocell so that photoelectrons are emitted. The maximum kinetic energy of photoelectrons reaching anode A depends on

- (1) the kind of metal making the cathode surface.
- (2) the voltage of the d.c. source.
- (3) the intensity of the monochromatic light used.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

At emission, $K_{\max} = hf - \phi$, where ϕ depends on the cathode metal. The positive anode then accelerates the electrons, adding eV to their kinetic energy. Thus $K_{\max}$ at the anode depends on the metal and supply voltage. Intensity changes the number emitted, not the maximum energy at fixed frequency.

Question 2.6

Aurora Borealis (Northern lights) are often observed in skies at high latitudes. When energetic electrons from outer space collide with the oxygen atoms in the upper atmosphere, the oxygen atoms are excited. The subsequent emission of light is usually green light of wavelength 558 nm. The minimum speed of these energetic electrons is of order of magnitude

- A. 10^2 m s^{-1} .
- B. 10^4 m s^{-1} .
- C. 10^6 m s^{-1} .
- D. 10^8 m s^{-1} .

Official answer: C

At least the photon energy hc/λ must be supplied in a collision. Setting $\frac{1}{2}m_{\text{e}}v^2 \approx hc/\lambda$ gives $v \approx \sqrt{[2(6.63 \times 10^{-34})(3.00 \times 10^8)/((9.11 \times 10^{-31})(558 \times 10^{-9}))]} \approx 8.85 \times 10^5 \text{ m s}^{-1}$. Its order of magnitude is 10^6 m s^{-1} .

Question 2.7

Which of the following can increase the resolving power of a transmission electron microscope (TEM)?

- (1) increasing the anode voltage in the electron gun
- (2) decreasing the aperture of the magnetic objective lens
- (3) increasing the separation between the magnetic projection lens and the fluorescent screen

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

Official answer: A

A higher accelerating voltage increases electron momentum and reduces $\lambda = h/p$, improving diffraction-limited resolution. A smaller objective aperture increases the minimum resolvable angle, worsening resolution.

Increasing projection distance can increase image size but does not improve the underlying resolving power.

Only (1).

Question 2.8

Zinc oxide (ZnO) is used in some suntan lotion. Which of the following statements is/are correct?

- (1) Nano-sized ZnO can block ultra-violet radiation while bulk-sized ZnO cannot.
- (2) Nano-sized ZnO is more effective in reflecting visible light than bulk-sized ZnO.
- (3) Suntan lotion with nano-sized ZnO appears transparent when applied to the skin.

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: B

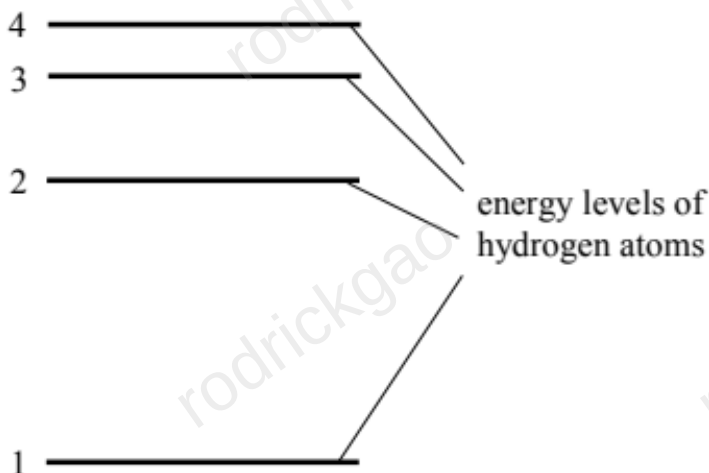
Both bulk and nano-sized ZnO can block ultraviolet radiation, so (1) is false. Nano-sized particles scatter visible light less strongly, not more, and the lotion can therefore appear transparent while retaining UV-blocking behaviour. Only (3) is correct.

Question 2 Structured Question

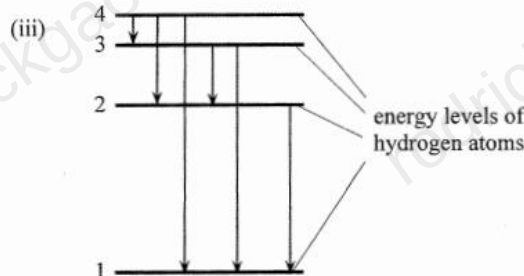
The energy levels E_n of an electron in a hydrogen atom from the Bohr model can take the form below:

$$E_n = -13.6/n^2 \text{ eV, where } n = 1, 2, 3, \dots$$

- (a) Bohr's idea was sometimes criticized by some physicists as semi-classical and semi-quantum. Point out ONE classical aspect in the Bohr model. (1 mark)
- (b) From the energy point of view, state the physical meaning of a hydrogen atom being in its ground state. (1 mark)
- (c) If the minimum energy required to ionize a hydrogen atom in the ground state is E , express the minimum momentum p of a photon for ionizing such a hydrogen atom in terms of E and another physical constant. (2 marks)
- (d) Hydrogen atoms in the ground state are bombarded by electrons each with kinetic energy 12.9 eV.
- (d)(i) Show that these hydrogen atoms can be excited at most to the third excited state (i.e. $n = 4$). (2 marks)
- (d)(ii) For a hydrogen atom excited to the third excited state ($n = 4$), what is the de Broglie wavelength of the orbiting electron in the atom? Given: the orbital radius r_n of the electron in a hydrogen atom from the Bohr model is equal to $0.053 n^2$ (unit: nm), where $n = 1, 2, 3, \dots$ (2 marks)
- (d)(iii) Copy the energy level diagram below to your answer book and draw arrow(s) to illustrate all possible transitions leading to emission of photons by these excited hydrogen atoms. (2 marks)



Official marking scheme

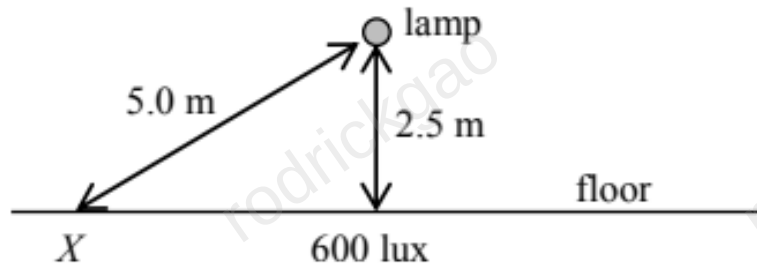
Solution	Marks	Remarks
2. (a) - the electron is considered as a particle revolving around the nucleus in definite orbits/circular motion; or - the centripetal force is provided by the Coulomb force; or - the electron's motion obeys Newton's laws of motion	1A	
	1	
(b) lowest energy level <u>or</u> most stable state	1A	
	1	
(c) $p = \frac{h}{\lambda} = \frac{hc}{\lambda} \cdot \frac{1}{c}$ $p = \frac{E}{c}$	1M	
	1A	
	2	
(d) (i) $E_4 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$, $\Delta E_{1 \rightarrow 4} = E_4 - E_1 = -0.85 - (-13.6) = 12.75 \text{ eV}$ $E_5 = -\frac{13.6}{5^2} = -0.544 \text{ eV}$, $\Delta E_{1 \rightarrow 5} = E_5 - E_1 = -0.544 - (-13.6) = 13.056 \text{ eV}$ $12.75 \text{ eV} < 12.9 \text{ eV} < 13.06 \text{ eV}$, therefore at most to the 3 rd excited state ($n = 4$).	1M	
Alternatively: $\Delta E = E_n - E_1 = -13.6 \left(\frac{1}{n^2} - \frac{1}{1^2} \right) = 12.9 \text{ eV}$ $n = 4.41$ and as n is an integer, it can at most take $n = 4$ (3 rd excited state).	1M	
	1A	
	2	
(ii) $mvr_n = \frac{nh}{2\pi} \Rightarrow 2\pi r_n = \frac{nh}{mv} = n\lambda$ (from postulate) When $n = 4$, $2\pi(0.053)(4^2) = 4\lambda$ Therefore, $\lambda = 1.33 \text{ nm}$	1M	
	1A	
Alternatively: $r = (0.053) 4^2 \text{ nm} = 0.848 \text{ nm} = 8.48 \times 10^{-10} \text{ m}$ $\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r^2} = \frac{mv^2}{r} \Rightarrow v^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{r} \cdot \frac{1}{m} = 9 \times 10^9 \times \frac{(1.6 \times 10^{-19})^2}{8.48 \times 10^{-10}} \cdot \frac{1}{9.11 \times 10^{-31}}$ $\Rightarrow v = 5.46 \times 10^5 \text{ m s}^{-1}$ $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(5.46 \times 10^5)} = 1.33 \times 10^{-9} \text{ m} = 1.33 \text{ nm}$	1M	
	1A	
	2	
(iii) 	2A	
	2	

- (a) One classical aspect: the electron follows a definite circular orbit around the nucleus, with Coulomb attraction providing the centripetal force according to Newton's laws.
- (b) The atom is in its lowest possible energy state, $n = 1$.
- (c) For a photon, $E = hf = hc/\lambda$ and $p = h/\lambda$. Therefore the minimum photon momentum is $p = E/c$. Do not use the non-relativistic massive-particle relation $E = p^2/(2m)$.
- (d)(i) $E_4 = -13.6/4^2 = -0.850$ eV, so excitation from $n = 1$ requires 12.750 eV. $E_5 = -13.6/5^2 = -0.544$ eV, requiring 13.056 eV. Since $12.750 < 12.9 < 13.056$, $n = 4$ is reachable but $n = 5$ is not.
- (d)(ii) The standing-wave condition is $2\pi r_n = n\lambda$. At $n = 4$, $r_4 = 0.053(4^2) = 0.848$ nm, so $\lambda = 2\pi(0.848)/4 = 1.33$ nm.
- (d)(iii) Draw all six downward arrows: $4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 1$, and $2 \rightarrow 1$. Include cascades through intermediate levels, not just the three direct transitions from $n = 4$.

Section C Energy and Use of Energy

Question 3.1

A lamp is fixed on the ceiling of a room as the only light source. The illuminance on the floor directly under the lamp is 600 lux. Assuming that the lamp emits light uniformly in all directions, what is the illuminance on the floor around point X as indicated in the diagram? Neglect any reflections of the walls and the ceiling.



- A. 75 lux
- B. 130 lux
- C. 150 lux
- D. 300 lux

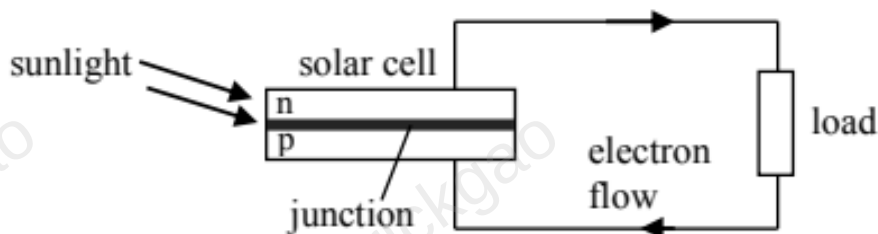
Official answer: A

The distance to X is 5.0 m instead of 2.5 m, and $\cos \theta = 2.5/5.0 = 0.5$ for incidence on the horizontal floor.

Illuminance varies as $\cos \theta / r^2$. Thus $E_X = 600(2.5/5.0)^2(0.5) = 75$ lux. Include both inverse-square spreading and oblique incidence.

Question 3.2

The schematic diagram below shows a solar cell under sunlight. Incident light photons reach the cell's junction of p-type and n-type semiconductor layers. Which of the following sequences correctly explains how electrical energy can be supplied to the load?



- (1) The electric field developed within the junction draws free electrons and holes to the n-type and p-type layers respectively.
- (2) Free electrons flow to the p-type layer through the load and rejoin the holes there.
- (3) Incident light photons knock out electrons from atoms in the junction such that free electrons result, leaving the holes behind.

- A. (1) → (2) → (3)
 B. (1) → (3) → (2)
 C. (3) → (2) → (1)
 D. (3) → (1) → (2)

Official answer: D

First, absorbed photons create electron-hole pairs (3). The junction's electric field separates them (1). Electrons then flow through the external load and recombine with holes on the p-side (2), delivering electrical energy. The sequence is (3) → (1) → (2).

Question 3.3

A satellite is powered by a solar panel of an area of 100 m^2 and having a conversion efficiency of 15%. Sunlight falls from a direction 30° to the normal of the panel. The solar constant is 1370 W m^{-2} . Estimate the electrical power output of this solar panel.

- A. 10.3 kW
 B. 17.8 kW
 C. 20.6 kW
 D. 58.2 kW

Official answer: B

The effective area perpendicular to the sunlight is $A \cos 30^\circ$. Electrical output = $\eta I A \cos 30^\circ = 0.15(1370)(100) \cos 30^\circ \approx 1.78 \times 10^4 \text{ W} = 17.8 \text{ kW}$. The given angle is to the normal, so use cosine.

Question 3.4

A coal-fired power plant generates and transmits electrical energy to consumers a long distance away. When a consumer connects an incandescent lamp to the mains supply, the following data illustrates what happens per 1000 J of energy in coal converted to electrical energy which is then supplied to the lamp.

Energy transfer	Energy
Loss during generation process in the power plant	600 J
Loss during transmission before reaching the lamp	100 J
Heat generated by the lamp	250 J
Visible light generated by the lamp	50 J

What is the end-use energy efficiency of this incandescent lamp?

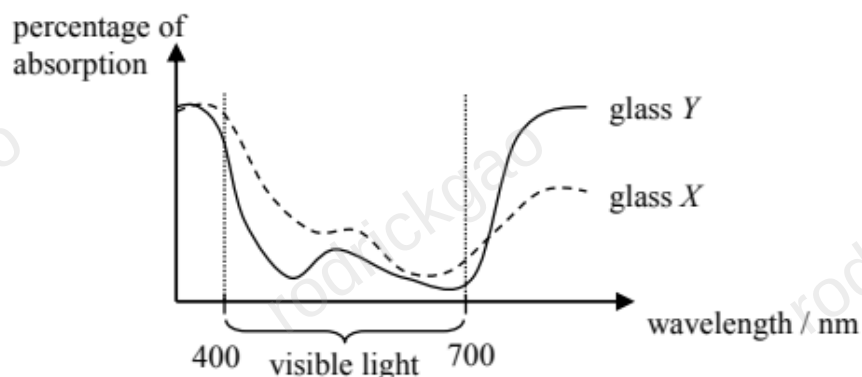
- A. 5%
- B. 12.5%
- C. 16.7%
- D. 30%

Official answer: C

End-use efficiency refers to useful output divided by the energy actually supplied to the lamp. The lamp receives $250 + 50 = 300$ J and gives 50 J of visible light. Hence efficiency = $(50/300) \times 100\% = 16.7\%$.
Dividing by the initial coal energy would give overall efficiency, not end-use efficiency.

Question 3.5

The following graph shows the percentage of absorption of different electromagnetic radiations when passing through two types of glasses X and Y. One of them is to be chosen for windows of buildings in Hong Kong. With the consideration of energy efficiency of buildings, which statement is correct?



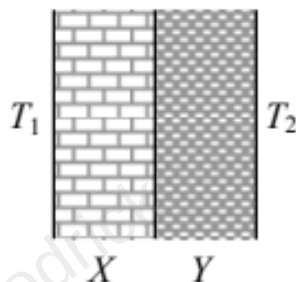
- A. X should be chosen as it is better than Y in reducing the need of air conditioning and the amount of lighting.
- B. Y should be chosen as it is better than X in reducing the need of air conditioning and the amount of lighting.**
- C. X should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than Y.
- D. Y should be chosen as it can greatly reduce the need of air conditioning although it needs a bit more amount of lighting than X.

Official answer: B

In the graph, Y absorbs more infrared radiation but less visible light than X. In the model used by this question, it transmits less solar infrared into the room while admitting more daylight, reducing both air-conditioning and artificial-lighting needs. Therefore Y is preferred for both reasons: B.

Question 3.6

The diagram shows a wall composed of two layers X and Y of equal thicknesses. The thermal conductivity of the material of X is higher than that of Y. The two sides of the wall are maintained at different temperatures T_1 and T_2 . Which of the following statements is/are correct?



- (1) The thermal transmittance of layer X is higher than that of layer Y.
- (2) The energy flowing through layer X per second is greater than that through layer Y.
- (3) The temperature drop across layer X is smaller than that across layer Y.

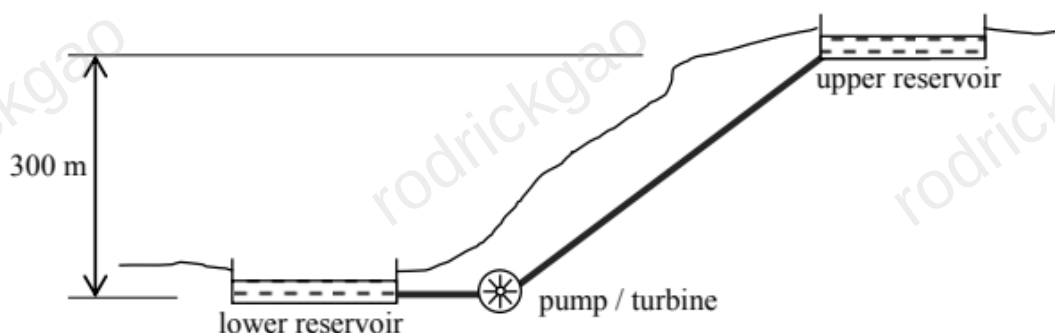
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

For equal thickness d , $U = \kappa/d$ is larger for X. In steady state, the heat-transfer rate is the same through both layers in series. From $P = \kappa A \Delta T/d$, a larger κ needs a smaller ΔT for the same P . Thus (1) and (3) are correct; (2) is false.

Question 3.7

A power plant is always running at its maximum output power of 1800 MW. However, as the demand of electricity fluctuates greatly within a day, a hydroelectric storage system is therefore designed to increase the output in the high demand period. During the 14 hours of low demand period in a day, the average demand is only 600 MW and the excess output is used to pump water from the lower reservoir to the upper one. When the demand is higher than 1800 MW, the water returns to the lower reservoir through a turbine to generate electricity.



Assume that the hydroelectric storage system is 100% efficient, find the minimum capacity (in kg of water) of the upper reservoir required. ($g = 9.81 \text{ m s}^{-2}$)

- A. $1.03 \times 10^{10} \text{ kg}$
- B. $1.47 \times 10^{10} \text{ kg}$
- C. $2.06 \times 10^{10} \text{ kg}$
- D. $3.08 \times 10^{10} \text{ kg}$

Official answer: C

The surplus power is $1800 - 600 = 1200 \text{ MW}$, stored for 14 h. Equate stored energy to mgh : $m = (1200 \times 10^6)(14 \times 3600)/(9.81 \times 300) \approx 2.06 \times 10^{10} \text{ kg}$. Only surplus output is stored, not the plant's full output.

Question 3.8

What is the function of the moderator in a nuclear fission reactor?

- A. It slows down neutrons and this helps increase the rate of nuclear fission.
- B. It slows down neutrons and this helps reduce the rate of nuclear fission.
- C. It absorbs neutrons and this helps reduce the rate of nuclear fission.
- D. It generates neutrons for nuclear fission.

Official answer: A

The moderator slows fast neutrons, making them more likely to cause further fission of fissile nuclei such as uranium-235 in a thermal reactor. Absorbing neutrons to regulate the chain reaction is the role of control rods, not the moderator.

Question 3 Structured Question

Figure 3.1 shows the Energy Efficiency Label of an air-conditioner X.

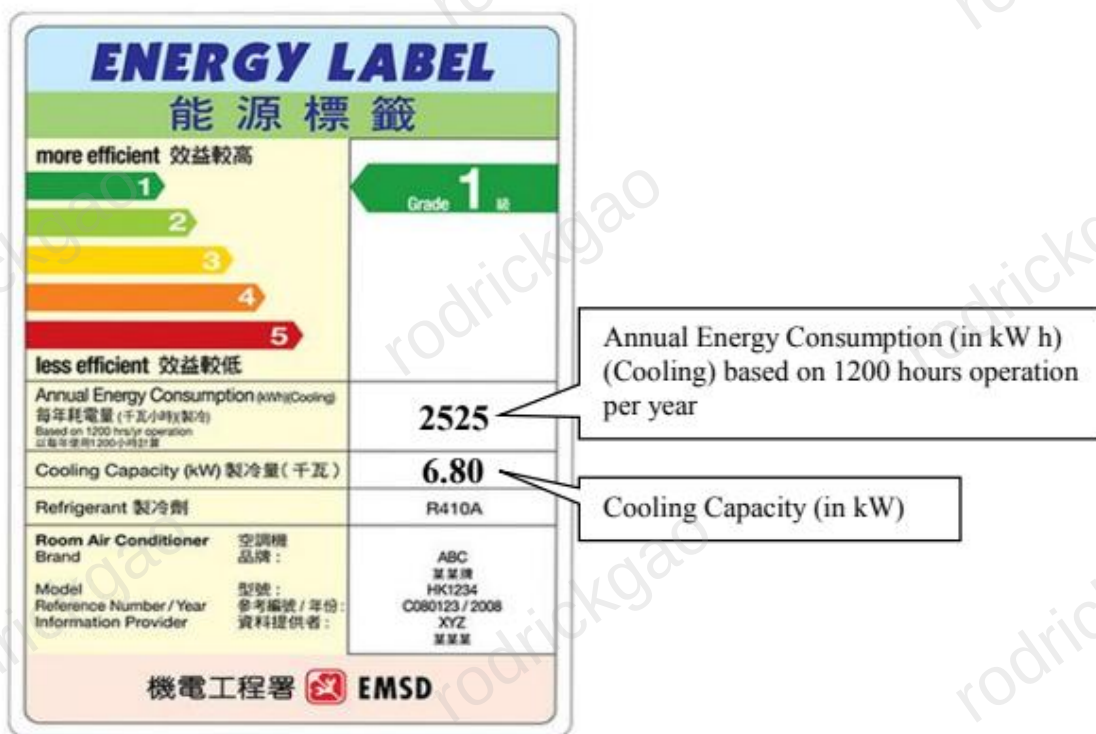


Figure 3.1

(a) This air-conditioner is installed in a room of floor area of 20.0 m^2 and height of 3.0 m .

Given: density of air = 1.2 kg m^{-3} , specific heat capacity of air = $1000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$.

(a)(i) Estimate the time required to cool the room from $33 \text{ }^\circ\text{C}$ to $25 \text{ }^\circ\text{C}$. Assume that the density and specific heat capacity of air remain constant throughout. (2 marks)

(a)(ii) Suggest a reason why the actual time for cooling from $33 \text{ }^\circ\text{C}$ to $25 \text{ }^\circ\text{C}$ is longer than the result found in (a)(i). (1 mark)

(b)(i) Find the average electrical power input (in kW) of air-conditioner X during operation. (1 mark)

(b)(ii) Find the value of cooling capacity / electrical power input for this air-conditioner. A student comments that this ratio having a value greater than 1 violates the principle of conservation of energy because the amount of heat removed by the air-conditioner is greater than the electrical power input. Discuss the student's comment. (3 marks)

(c) X is a cool-only air-conditioner as it can only cool air. Nowadays, "reverse-cycle air-conditioners" (RCAC) that can either cool or warm air are available in the market and they have the same major components of X. Figure 3.2 shows a simplified schematic diagram of an RCAC with four components A, B, C and D, in which A is the expansion valve and C is the compressor.

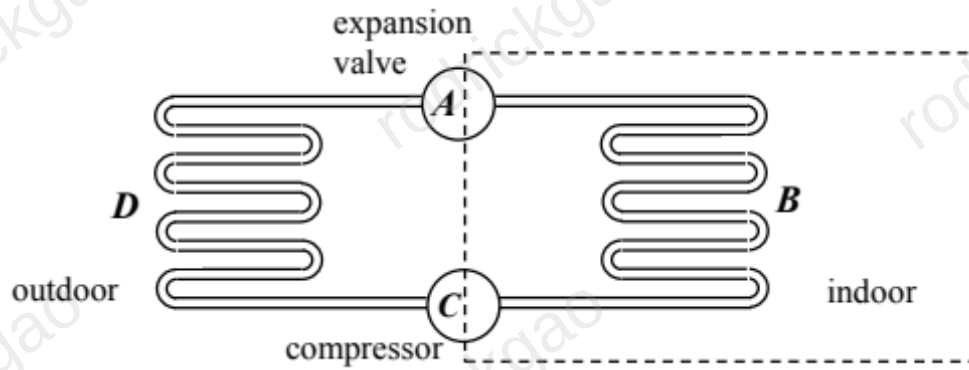


Figure 3.2

(c)(i) If the RCAC operates in a mode to warm air during winter time, give the direction of the flow of refrigerant starting from compressor C using the letters A, B, C and D. State in which component, A, B or D, the refrigerant has the highest temperature. (2 marks)

(c)(ii) Suggest ONE modification that can convert a cool-only air-conditioner into an RCAC. (1 mark)

Official marking scheme

Solution		Marks	Remarks
3. (a) (i)	Time required = $\frac{\text{heat to remove } (mc\Delta T)}{\text{cooling capacity}}$	1M	
	$= \frac{[(20.0 \times 3.0) \times 1.2] \times 1000 \times (33 - 25)}{6.80 \times 1000}$		
	$= \frac{576000}{6800} = 85 \text{ s (1.42 min. or 0.0236 hr)}$	1A	
		2	
	(ii) ANY ONE: Heat has to be removed from wall, furniture etc. / Heat transferred from outside to the room has to be removed / Other reasonable factors such as bad air ventilation for air-conditioner / Doors or windows are not closed properly / Installation position is facing west or exposed to sunlight directly, etc. / Heat gain from surroundings / Poor conductor (air) lengthens the time for heat transfer	1A	
		1	
	(b) (i) $P_{\text{in}} = \frac{2525}{1200} = 2.1 \text{ (kW) or } 2100 \text{ W}$	1A	
		1	
	(ii) $\frac{\text{cooling capacity}}{\text{electrical power input}} (\text{COP}) = \frac{6.80}{2.1} = 3.24$ Conservation of energy is not violated. For each joule of electrical energy consumed by the air-conditioner/compressor, 3.24 J of heat will be transferred/removed, but not created, by the air-conditioner.	1M/1A	
		1A	
(c) (i)	$(C \rightarrow) B \rightarrow A \rightarrow D$ component B (or condenser)	1A	
		1A	
		2	
	(ii) reverse the direction of flow of refrigerant	1A	
	Or interchange/swap the positions of B (condenser) and D (evaporator) or A (expansion valve) and C (compressor)		
		1	

(a)(i) Air mass $m = \rho V = 1.2(20.0 \times 3.0) = 72 \text{ kg}$. Heat to remove $Q = mc\Delta T = 72(1000)(33 - 25) = 576000 \text{ J}$.
Time = $Q/(\text{cooling capacity}) = 576000/6800 = 84.7 \text{ s} \approx 85 \text{ s}$.

(a)(ii) For example, heat must also be removed from the walls and furniture. This extra thermal load was excluded from the air-only estimate.

(b)(i) Average electrical input = $2525 \text{ kW h} / 1200 \text{ h} = 2.104\dots \text{ kW} \approx 2.1 \text{ kW}$.

(b)(ii) Ratio (COP) = $6.80/(2525/1200) \approx 3.23$; using the rounded 2.1 kW gives 3.24 , as in the official scheme. It does not violate conservation of energy: the machine transfers heat from the room instead of creating that heat from electrical energy.

(b)(ii) Over the same time interval, heat rejected outdoors Q_H = heat removed indoors Q_C + electrical work W . Thus Q_C/W may exceed 1 while the complete energy balance remains satisfied.

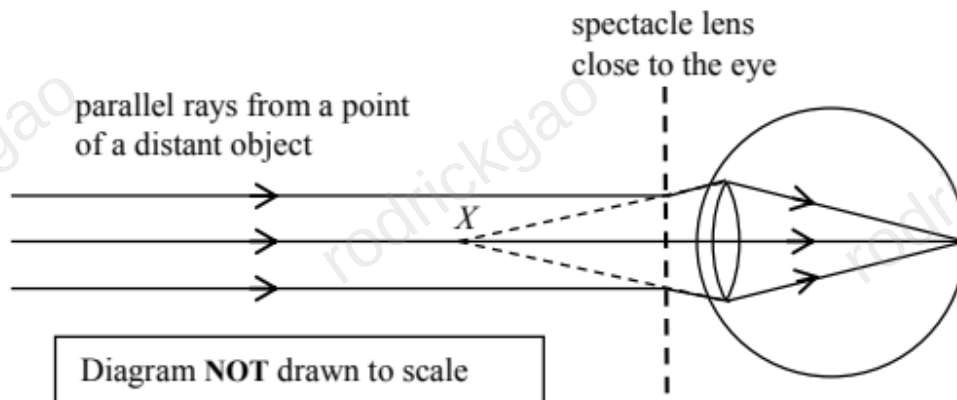
(c)(i) Flow is $C \rightarrow B \rightarrow A \rightarrow D \rightarrow C$. B is the hot indoor condenser and has the highest refrigerant temperature among A, B and D.

(c)(ii) Add a reversing arrangement that reverses refrigerant circulation between the indoor and outdoor heat exchangers, so their condenser and evaporator roles can be exchanged.

Section D Medical Physics

Question 4.1

Peter suffers from a certain eye defect and needs to wear spectacles. The diagram shows the spectacle lens for correcting this defect. Which statements are correct?



- (1) Peter is suffering from short sightedness.
- (2) Point X is the near point of his unaided eye.
- (3) If X is 0.8 m from the spectacle lens, the power of the lens should be -1.25 D.

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

Official answer: B

The diverging spectacle lens forms a virtual image of a distant object at X, the unaided eye's far point, correcting short sight. Thus (1) is true and (2) is false. For parallel incident rays, $f = -0.8$ m, so power = $1/f = -1.25$ D. Hence (1) and (3).

Question 4.2

Which of the following are disadvantages of a fibre optic endoscope?

- (1) Anesthesia is sometimes required.
- (2) It can only be used for viewing the inner surface of an organ with cavity.
- (3) It may cause internal bleeding.

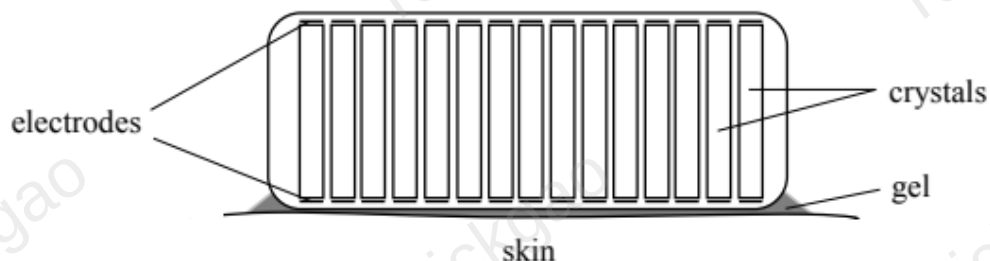
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

Official answer: D

In the fibre-optic endoscopy context of this question, all three are limitations: insertion may require anaesthesia, the instrument provides a surface view inside an accessible cavity rather than a cross-sectional image through tissue, and insertion may injure tissue and cause bleeding. The official answer is D.

Question 4.3

The following diagram shows an ultrasound scanner with an array of crystals. A pair of electrodes is connected across each crystal. Which of the following statements is/are correct?



- (1) The gel is used to reduce the attenuation of ultrasound when passing the gap between the scanner and the skin.
- (2) When a crystal receives ultrasound, an electrical signal is generated between the electrodes by piezoelectric effect.
- (3) This scanner is designed for taking B-scan images.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only**

Official answer: D

A receiving piezoelectric crystal converts mechanical deformation into an electrical signal, and the array supplies spatially separated echo information for a B-scan image. Thus (2) and (3) are correct. For (1), the gel's intended role is to remove the air gap and reduce reflection caused by acoustic-impedance mismatch, not to reduce propagation attenuation within the gap. This distinction is why the official answer is D.

Question 4.4

Ultrasound scanning is NOT suitable for the lungs because air in the lungs

- A. has a high attenuation coefficient and can absorb almost all the ultrasound.
- B. has a low attenuation coefficient and can absorb almost all the ultrasound.
- C. has high acoustic impedance compared to soft tissue and can reflect almost all the ultrasound.
- D. has low acoustic impedance compared to soft tissue and can reflect almost all the ultrasound.**

Official answer: D

Air has a much lower acoustic impedance $Z = \rho c$ than soft tissue. The large mismatch makes the reflected fraction $[(Z_{\text{tissue}} - Z_{\text{air}})/(Z_{\text{tissue}} + Z_{\text{air}})]^2$ close to 1, so very little ultrasound enters across a tissue-air boundary. The intended reason is strong reflection, not almost complete absorption.

Question 4.5

The diagram shows an X-ray beam passing through a metal block of linear attenuation coefficient μ . Its intensity is reduced from I to $I/2$. If the metal block is replaced by another one of the same thickness but having a linear attenuation coefficient of 3μ , what would the intensity of the emergent X-ray beam be?

- A. $I/6$
- B. $I/8$
- C. $I/9$
- D. $I/16$

Official answer: B

For thickness x , $I_{\text{out}} = I e^{-\mu x}$. The first block gives $e^{-\mu x} = 1/2$. For coefficient 3μ at the same thickness, $I_{\text{out}} = I e^{-3\mu x} = I(e^{-\mu x})^3 = I(1/2)^3 = I/8$. Attenuation is exponential, not linear.

Question 4.6

The image of a computed tomography (CT) scan is of size $32 \text{ cm} \times 32 \text{ cm}$ and the size of each pixel of the image is 0.391 mm^2 . Which matrix size below corresponds to the resolution of this CT scan image?

- A. 128×128
- B. 256×256
- C. 512×512
- D. 1024×1024

Official answer: C

The pixel area is 0.391 mm^2 , so a square pixel has side $\sqrt{0.391} \approx 0.625 \text{ mm}$. The image side is 320 mm , giving about $320/0.625 = 512$ pixels along each side. Thus the matrix is 512×512 . Do not treat the pixel area as a length.

Question 4.7

A doctor decides to conduct radionuclide imaging of a patient's kidneys. Which of the following is the main reason for choosing radionuclide imaging rather than other imaging methods?

- A. Diagnosis of the kidneys' function can be made.
- B. Fine details of the kidneys' structure can be seen.
- C. The image produced has the highest resolution compared to the other methods.
- D. Diagnosis will give highly specific information of the kind of kidney disease.

Official answer: A

Radionuclide imaging follows the uptake and clearance of a tracer, providing information about organ function. Its main advantage here is functional assessment, not the finest structural detail or the highest spatial resolution. Therefore A.

Question 4.8

The following radioactive sources are all non-toxic and readily absorbed by a certain organ. Which one is the most suitable to be used as a tracer for radionuclide imaging of that organ?

- A. a γ -source of half-life 16 hours
- B. a γ -source of half-life 8 months
- C. a β -source of half-life 20 seconds
- D. a β -source of half-life 12 hours

Official answer: A

Gamma radiation can escape the body for external detection. A half-life of 16 hours allows the examination while avoiding the prolonged activity of an 8-month source. Beta radiation is less suitable for external imaging, and 20 seconds is too short for normal tracer handling and uptake. Thus A is the best of the listed sources.

Question 4 Structured Question

Figure 4.1 shows the structure of human ear.

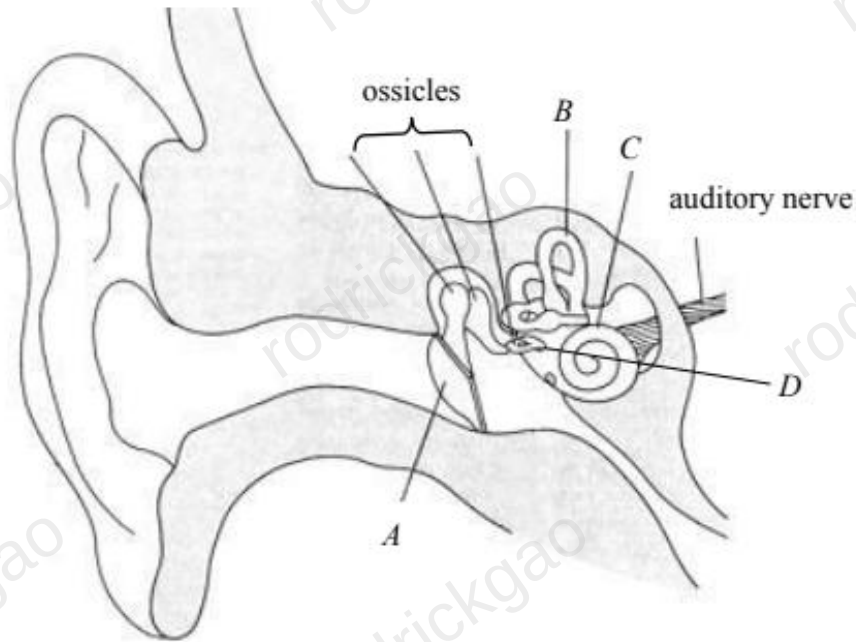


Figure 4.1

(a)(i) Match the letters A, B, C and D in Figure 4.1 with the parts of the ear, namely, eardrum, oval window, semi-circular canals and cochlea. State the function of C. (2 marks)

(a)(ii) The ratio of area of A to that of D is 20. If the ear amplifies the pressure of a sound signal by 25 times totally after passing D, find the gain of pressure contributed by the lever action of the ossicles. (1 mark)

(b) Figure 4.2 shows an equal loudness curve of people with normal hearing.

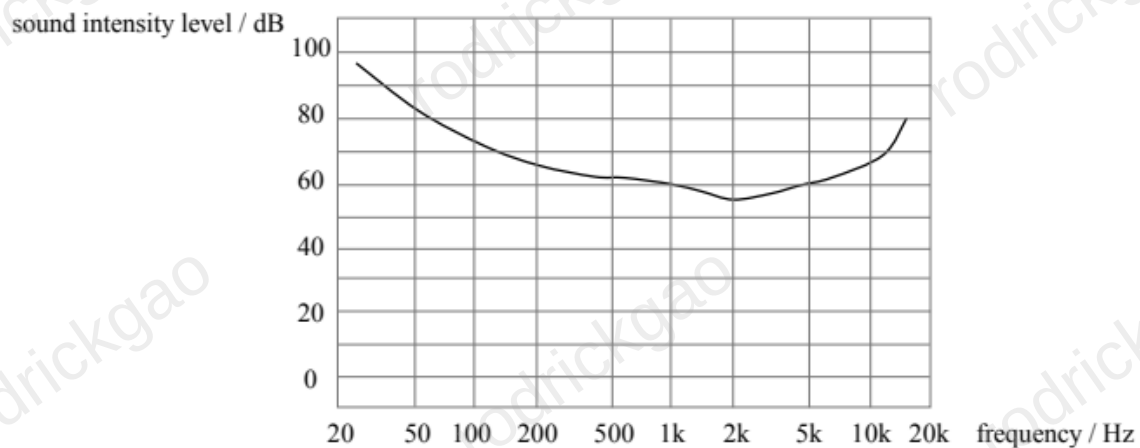


Figure 4.2

(b)(i) State the loudness, in phons, represented by this curve. State the physical significance of the curve being higher up at both ends. (2 marks)

(b)(ii) A worker has suffered from hearing loss due to prolonged exposure to a noisy environment and the loss is most severe for sounds around kHz frequencies. If the worker is tested for the threshold of hearing, which of the equal loudness curves, A, B and C, in Figure 4.3 best represents his response? Explain your choice. (2 marks)

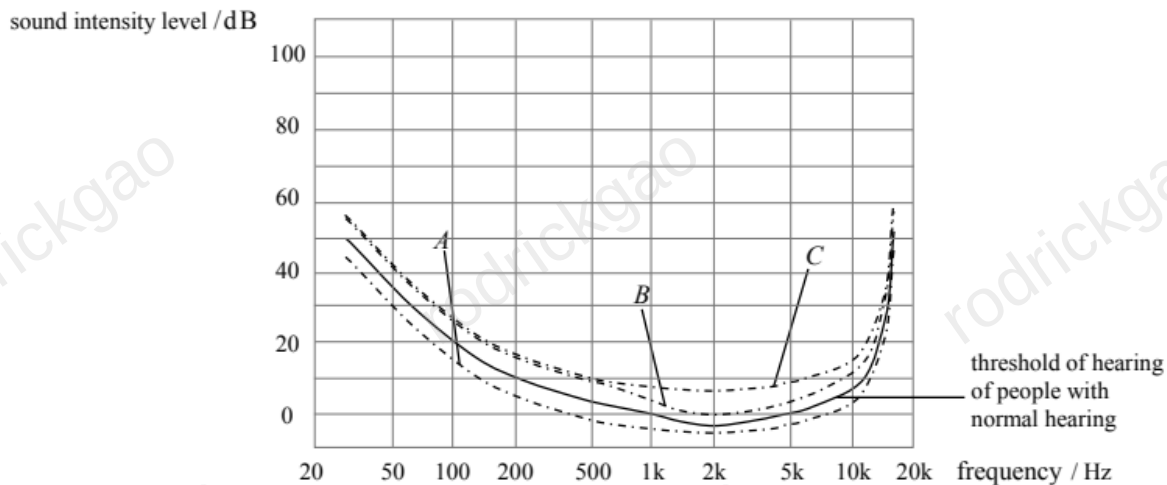


Figure 4.3

(c) An engineer working near an operating engine is exposed to noise of sound intensity 80 W m^{-2} . After putting on earmuffs, the sound intensity of the noise heard is reduced to $2.5 \times 10^{-5} \text{ W m}^{-2}$. Estimate the decrease in sound intensity level of the noise, in dB, heard after wearing the earmuffs. (3 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) (i) A: eardrum B: semi-circular canals C: cochlea D: oval window C (cochlea) is for discriminating different frequencies of incoming sound waves / convert sound waves to nerve signals / auditory sensor cells inside send signals to brain.	1A 1A	
	2	
(ii) $25 \div 20 = 1.25$ (i.e. 25% increase)	1M/1A	
	1	
(b) (i) 60 (phons) The ear is less sensitive (compared to 1~2 kHz frequency) to sound of low or high frequencies / more sensitive to middle frequencies / need a higher sound intensity to give the same loudness at high and low frequencies.	1A 1A	
	2	
(ii) Curve C. Curve shifted upwards such that a greater intensity level for threshold of hearing (or giving the same loudness sensation), especially significant in kHz range.	1A 1A	
	2	
(c) Change in sound intensity level $L_1 = 10 \log \frac{80}{I_0}$ $L_2 = 10 \log \frac{2.5 \times 10^{-5}}{I_0}$ $L_2 - L_1 = 10 \log \frac{80}{2.5 \times 10^{-5}}$ $= -65 \text{ (dB)}$	1M 1M	
Alternatively: Assume $I_0 = 10^{-12} \text{ W m}^{-2}$ $L_1 = 10 \log \frac{80}{10^{-12}} = 139.03 \text{ dB}$ $L_2 = 10 \log \frac{2.5 \times 10^{-5}}{10^{-12}} = 74.03 \text{ dB}$ $L_2 - L_1 = -65 \text{ (dB)}$ Or $10 \log \left(\frac{I_{\text{noise reduced}}}{I_{\text{original}}} \right)$ $= 10 \log \left(\frac{2.5 \times 10^{-5}}{80} \right)$ $= -65 \text{ dB}$ $\therefore \text{reduced by } 65 \text{ (dB)}$	1M 1M+1A 2M 1A	Accept $\pm 65 \text{ dB}$
	3	

(a)(i) A: eardrum. B: semi-circular canals. C: cochlea. D: oval window. The cochlea distinguishes sound frequencies and converts mechanical vibrations into nerve signals through its sensory cells.

(a)(ii) Total pressure amplification = area-ratio gain $\times$ lever gain. Hence lever gain = $25/20 = 1.25$ times, equivalent to a 25% increase.

(b)(i) At 1 kHz the curve has intensity level 60 dB, so its loudness level is 60 phons. The higher ends show that the ear is less sensitive to low and high frequencies: a higher intensity is needed for the same perceived loudness.

(b)(ii) Curve C. Hearing loss raises the threshold intensity needed for detection, with the greatest upward departure from the normal threshold in the kHz range.

(c) Decrease = $L_{\text{before}} - L_{\text{after}} = 10 \log_{10}(I_{\text{before}}/I_{\text{after}}) = 10 \log_{10}[80/(2.5 \times 10^{-5})] = 65.05... \text{ dB} \approx 65 \text{ dB}$. If reporting change as $L_{\text{after}} - L_{\text{before}}$, it is -65 dB ; the requested decrease is positive 65 dB.